

Policy Gradient Methods

Policy Gradient

- Learn a parametrized policy
- Don't need a value function to pick an action

$$\pi(a|s, \theta) = P[A_t = a | S_t = s, \theta]$$

- Value function may still be used to update the policy parameter
 - The performance measure we would like to maximize

Policy Gradient

- Performance measure $J(\theta)$

- The gradient ascent step is

$$\theta_{t+1} = \theta_t + \alpha \widehat{\nabla J(\theta_t)}$$

- As in the gradient bandit algo, we only have noisy estimates of $J(\theta_t)$
- A desirable property is ...

Example soft-max policy parametrization

$$\pi(a|s, \theta) = \frac{e^{h(s, a, \theta)}}{\sum_a e^{h(s, a, \theta)}}$$

- $h(s, a, \theta)$ is a numerical preference for the state-action pair
 - Could be a neural net
- Soft-max allows the policy to approach a deterministic policy
- We require the policy function to be differentiable wrt θ

What All Do We Need>

- A differentiable policy parametrization
- A performance measure that we want to maximize
- A way to calculate its gradient
 - Given by the Policy Gradient Theorem
- In practice, we calculate the gradient of a (noisy) estimate of the performance measure

Performance Measure We Want To Maximize

$$\sum_s v_{\pi_{\theta}}(s) P[S_0=s]$$

Value function

$$J(\theta) \triangleq v_{\pi_{\theta}}(s)$$

Policy Gradient Theorem tells us how to calculate the gradient

$$\nabla J(\theta) \propto \sum_s \mu(s) \sum_a \nabla \pi_{\theta}(a|s) q_{\pi_{\theta}}(s, a)$$

Proof of the Policy Gradient Theorem

$$J(\theta) = V_{\pi_{\theta}}(s) = \sum_a q_{\pi_{\theta}}(s, a) \pi_{\theta}(a|s)$$

$$\nabla_{\theta} V_{\pi_{\theta}}(s) = \sum_a \left(\underbrace{\nabla_{\theta} q_{\pi_{\theta}}(s, a)}_J \pi_{\theta}(a|s) + \nabla_{\theta} \pi_{\theta}(a|s) q_{\pi_{\theta}}(s, a) \right)$$

We have:

$$\nabla J(\theta) = \sum_a q_{\pi_\theta}(s,a) \nabla \pi_\theta(a|s) + \sum_a \nabla q_{\pi_\theta}(s,a) \pi_\theta(a|s)$$

Show that:

$$\nabla J(\theta) = \nabla v_{\pi_\theta}(s) = \sum_x \sum_{k=0}^{\infty} P[s \rightarrow x, k, \pi_\theta] \left(\sum_a \nabla \pi_\theta(a|x) q_{\pi_\theta}(x,a) \right)$$

$P[s \rightarrow x, k, \pi_\theta]$ is the k -step transition probability.

$$P[s \rightarrow x, 0, \pi_\theta] = \begin{cases} 1 & s=x \\ 0 & \text{otherwise} \end{cases}$$

$$\nabla J(\theta) = \underbrace{\sum_a q_{\pi_\theta}(s,a) \nabla \pi_\theta(a|s)}_A + \underbrace{\sum_a \nabla q_{\pi_\theta}(s,a) \pi_\theta(a|s)}_B$$

$$\nabla q_{\pi_\theta}(s,a) = \nabla \left(\sum_{s'} \sum_{a'} \left(\gamma + \sum_{a''} q_{\pi_\theta}(s',a'') \pi_\theta(a''|s') \right) P(s', \pi | s, a) \right)$$

$$= \nabla \left(\sum_{s'} \left(\sum_{a'} q_{\pi_\theta}(s',a') \pi_\theta(a'|s') \right) P(s'|s,a) \right) \leftarrow$$

$$= \sum_{s'} \nabla \left(\sum_{a'} \left(q_{\pi_\theta}(s',a') \pi_\theta(a'|s') \right) \right) P(s'|s,a)$$

$$= \sum_{s'} \sum_{a'} \left(\nabla q_{\pi_\theta}(s',a') \pi_\theta(a'|s') + \nabla \pi_\theta(a'|s') q_{\pi_\theta}(s',a') \right) P(s'|s,a)$$

$$= \sum_{s'} \sum_{a'} \left(\nabla \sum_{s''} \sum_{a''} q_{\pi_\theta}(s'',a'') \pi_\theta(a''|s'') P(s''|s',a') \pi_\theta(a'|s') P(s'|s,a) \right. \\ \left. + \sum_{s''} \sum_{a''} \nabla \pi_\theta(a''|s') q_{\pi_\theta}(s'',a'') P(s'|s,a) \right)$$

$$= \sum_{s'} \nabla \sum_{s''} \sum_{a''} q_{\pi_\theta}(s'',a'') \pi_\theta(a''|s'') P(s''|s') P(s'|s,a) \\ + \sum_{s'} \sum_{a'} \nabla \pi_\theta(a'|s') q_{\pi_\theta}(s',a') P(s'|s,a)$$

$$\begin{aligned} &P(s''|s',a') P(s'|s,a) \\ &= P(s''|s',a',s) P(s'|s,a) \end{aligned}$$

$$\sum_x \sum_{k=0}^{\infty} P[s \rightarrow x, k, \pi_\theta] \left(\sum_a \nabla \pi_\theta(a|x) q_{\pi_\theta}(x,a) \right)$$

$$= \underbrace{\sum_a \nabla \pi_\theta(a|s) q_{\pi_\theta}(s,a)}_{\text{Term A}}$$

(Also, the first term (for $k=0$) is final expression)

$\left\{ \begin{aligned} &+ \sum_x P[s \rightarrow x, 1, \pi_\theta] \left(\sum_a \nabla \pi_\theta(a|x) q_{\pi_\theta}(x,a) \right) \\ &\vdots \\ &\rightarrow \text{Term B} \end{aligned} \right.$

$$\sum_a \nabla q_{\pi_\theta}(s, a) \underbrace{\pi_\theta(a|s)}_{\text{(2nd term)}}$$

$$= \sum_a \left(\underbrace{\sum_{s'} \sum_{a'} \left(\nabla \sum_{s''} \sum_{a''} q_{\pi_\theta}(s'', a'') \pi_\theta(a''|s'') \underbrace{p(s''|s', a')} \underbrace{\pi_\theta(a'|s')} \underbrace{p(s'|s, a)} \right)}_{\text{1st term}} \right. \\ \left. + \underbrace{\sum_{s'} \sum_{a'} \nabla \pi_\theta(a'|s') q_{\pi_\theta}(s', a') p(s'|s, a)}_{\text{2nd term}} \right) \pi_\theta(a|s)$$

$$= \sum_a \left(\underbrace{\sum_{s'} \sum_{a'} \left(\nabla \sum_{s''} \sum_{a''} q_{\pi_\theta}(s'', a'') \pi_\theta(a''|s'') \underbrace{p(s''|s', a')} \underbrace{\pi_\theta(a'|s')} \underbrace{p(s'|s, a)} \right)}_{\text{1st term}} \pi_\theta(a|s) \right. \\ \left. + \sum_{s'} \sum_{a'} \nabla \pi_\theta(a'|s') q_{\pi_\theta}(s', a') \underbrace{p(s'|s)}_{P[s \rightarrow s', k=1, \pi_\theta]} \right)$$

$$= \sum_a \left(\underbrace{\sum_{s'} \sum_{a'} \left(\nabla \sum_{s''} \sum_{a''} q_{\pi_\theta}(s'', a'') \pi_\theta(a''|s'') \underbrace{p(s''|s', a')} \underbrace{\pi_\theta(a'|s')} \underbrace{p(s'|s, a)} \right)}_{\text{1st term}} \pi_\theta(a|s) \right. \\ \left. + \underbrace{\sum_{s'} \sum_{a'} \nabla \pi_\theta(a'|s') q_{\pi_\theta}(s', a') P[s \rightarrow s', k=1, \pi_\theta]}_{\text{2nd term in my final expression}} \right)$$

We have:

$$\nabla J(\theta) = \nabla V_{\theta}(s) = \sum_x \sum_{k=0}^{\infty} P[S \rightarrow x, k, \pi_{\theta}] \left(\sum_a \nabla \pi_{\theta}(a|x) q_{\pi_{\theta}}(x,a) \right)$$

Show that:

$$(a) \nabla J(\theta) \propto \sum_x \mu(x) \left(\sum_a \nabla \pi_{\theta}(a|x) q_{\pi_{\theta}}(x,a) \right)$$

(b) Further write down the R.H.S as an expectation.

(a) Consider $\sum_{k=0}^{\infty} P[S \rightarrow x, k, \pi_{\theta}]$

Suppose X_k takes the value 1 when x occurs k transitions later.

$\sum_{k=0}^{\infty} X_k$ is the total no of times x

occurs, given we started in state s .

$$\begin{aligned} \therefore E\left[\sum_{k=0}^{\infty} X_k\right] &= \sum_{k=0}^{\infty} E[X_k] \\ &= \sum_{k=0}^{\infty} P[S \rightarrow x, k, \pi_{\theta}]. \end{aligned}$$

We (tried to) show that:

$$\begin{aligned} \nabla V_{\theta}(s) &= \sum_x \sum_{k=0}^{\infty} P[S \rightarrow x, k, \pi_{\theta}] \left(\sum_a \nabla \pi_{\theta}(a|x) q_{\pi_{\theta}}(x,a) \right) \\ &= \sum_x \underbrace{\left(\sum_{x'} \sum_{k=0}^{\infty} P[S \rightarrow x', k, \pi_{\theta}] \right)}_{\mu(x)} \underbrace{\left(\frac{\sum_{k=0}^{\infty} P[S \rightarrow x, k, \pi_{\theta}]}{\sum_{x'} \sum_{k=0}^{\infty} P[S \rightarrow x', k, \pi_{\theta}]} \right)}_{\mu(x)} \left(\sum_a \nabla \pi_{\theta}(a|x) q_{\pi_{\theta}}(x,a) \right) \\ &\propto \sum_x \mu(x) \left(\sum_a \nabla \pi_{\theta}(a|x) q_{\pi_{\theta}}(x,a) \right) \end{aligned}$$

$$\therefore \nabla V_{\theta}(s) \propto \sum_x \mu(x) \left(\sum_a \nabla \pi_{\theta}(a|x) q_{\pi_{\theta}}(x,a) \right)$$

$$\begin{aligned} (b) \quad &= \sum_x \sum_a \underbrace{\mu(x) \pi_{\theta}(a|x)}_{\pi_{\theta}(a|x)} \underbrace{\frac{\nabla \pi_{\theta}(a|x)}{\pi_{\theta}(a|x)} q_{\pi_{\theta}}(x,a)}_{\frac{\nabla \pi_{\theta}(a|x)}{\pi_{\theta}(a|x)} q_{\pi_{\theta}}(x,a)} \\ &= E_{S_t, A_t} \left[\frac{\nabla \pi_{\theta}(a|x)}{\pi_{\theta}(a|x)} q_{\pi_{\theta}}(x,a) \right] \\ &= E_{S_t, A_t} \left[\nabla \ln \pi_{\theta}(A_t|S_t) \underbrace{q_{\pi_{\theta}}(S_t, A_t)}_{\text{MC update}} \right] \end{aligned}$$

Approximate:

$$\widehat{\nabla V_{\theta}(s)} = \nabla \ln \pi_{\theta}(A_t|S_t) G_t, \text{ assuming MC update.}$$

Policy Gradient Theorem

$$\nabla J(\theta) \propto \sum_s \mu(s) \sum_a \nabla \pi_\theta(a|s) q_{\pi_\theta}(s, a)$$



- Calculation of the gradient of the performance measure requires calculation of gradient of only the parametrized policy PMF
- But not the gradient of distribution of states...
- Note that $J(.)$ depends on the same, however

Policy Gradient

$$\begin{aligned}\nabla J(\theta) &\propto \sum_s \mu(s) \sum_a \nabla \pi_\theta(a|s) q_{\pi_\theta}(s, a) \\ &= E_{S_t, A_t} \left[\frac{\nabla \pi_\theta(A_t, S_t)}{\pi_\theta(A_t, S_t)} q_{\pi_\theta}(S_t, A_t) \right]\end{aligned}$$

- Use sample estimate of the expectation...

REINFORCE (PG with MC based targets)

$$\theta_{t+1} = \theta_t + \alpha \gamma^t G_t \frac{\nabla \pi_{\theta}(A_t, S_t)}{\pi_{\theta}(A_t, S_t)}$$

REINFORCE With Baseline

$$\theta_{t+1} = \theta_t + \alpha \gamma^t (G_t - \hat{v}_{\pi_{\theta}}(S_t, \mathbf{w})) \frac{\nabla \pi_{\theta}(A_t, S_t)}{\pi_{\theta}(A_t, S_t)}$$

Actor Critic Method

$$\theta_{t+1} = \theta_t + \alpha \gamma^t (R_{t+1} + \hat{v}_{\pi_\theta}(S_{t+1}, \mathbf{w}) - \hat{v}_{\pi_\theta}(S_t, \mathbf{w})) \frac{\nabla \pi_\theta(A_t, S_t)}{\pi_\theta(A_t, S_t)}$$

Actor-Critic Method

- 1: Initialize $\mathbf{w}$ and θ
- 2: **while** true **do**
- 3: Initialize step $t = 0$ at beginning of a new episode
- 4: Initial state S_0 , action A_0 using initial policy, reward R_1
- 5: **for** step t in an episode **do**

$$\mathbf{w}_{t+1} = \mathbf{w}_t +$$

$$\alpha(R_{t+1} + \gamma \hat{v}(S_{t+1}, \mathbf{w}_t) - \hat{v}(S_t, \mathbf{w}_t)) \nabla_{\mathbf{w}_t} \hat{v}(S_t, \mathbf{w}_t)$$

$$\theta_{t+1} = \theta_t + \beta \gamma^t (R_{t+1} + \hat{v}_{\pi_{\theta}}(S_{t+1}, \mathbf{w}) - \hat{v}_{\pi_{\theta}}(S_t, \mathbf{w})) \frac{\nabla \pi_{\theta}(A_t, S_t)}{\pi_{\theta}(A_t, S_t)}$$

- 6: Agent chooses A_{t+1} using $\pi_{\theta_{t+1}}$
 - 7: $t = t + 1$
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